MA Idea Description

1. Are you planning on working on the project you proposed in MACS 30200 in the Spring (if applicable). If yes, briefly note any changes you might have to make (e.g. adding another dataset, changing an analytical strategy after doing preliminary analysis, shifting your RQ etc). *Note: even if your plan has stayed completely the same, you are required to answer the rest of the questions.*

   Yes. Major changes are listed below:

   1) I plan to replace LIWC with BERT based sentiment and emotion classification due to licensing constraints.
   2) I added a null model validation suite to rigorously distinguish genuine emergent patterns from statistical artifacts. This suite consists of 3 tests, including permutation tests on the interaction network, temporal shuffling, and topic-interaction decoupling.
   3) I updated the timeline.
2. Are you planning on doing a professional or an academic thesis?

   Academic

3. What field or industry would your project best fit in? If writing an academic thesis, list 2-3 journals that feature the kind of research you plan to do. If writing a professional thesis, provide 2-4 links to the kinds of reports/white papers/etc. that are similar to what you plan to produce.

   Computational Social Science (Journals including Journal of Computational Social Science, Proceedings of the National Academy of Sciences)

4. What is your research question/goal? Why do you think this is a worthwhile question to pursue in your MA?

   RQ1 In a purely AI-driven simulated community, how do latent content themes and observable textual features predict different forms of social engagement?

   RQ2 What structural network patterns (specifically regarding homophily, echo chambers, and controversy) emerge from these algorithmically generated interactions?

   These are worthwhile questions to pursue because LLM agents are rapidly moving from isolated chatbot interactions into multi-agent environments where they autonomously form communities. Understanding whether these algorithmic communities replicate or systematically diverge from human social patterns has direct implications for platform governance, content moderation,

and our theoretical understanding of what counts as social behavior. The Moltbook, with 750K agent interactions, provides a uniquely large-scale, purely synthetic testbed that has not been systematically analyzed in the existing literature.

5. Identify at least five academic sources that are relevant to your work and write 2-3 sentences for each telling us why they're relevant.

**1. Binz, M., & Schulz, E. (2023). Using cognitive psychology to understand GPT-3. *PNAS*, 120(6).** This study demonstrates that LLM agents map onto classical cognitive psychological models, with responses on canonical tasks resembling human behavioral patterns. It provides the micro-level theoretical foundation for treating AI agent interactions as meaningful social behavior, which is the premise underlying my entire analysis of Moltbook.

**2. Park, J. S., et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior.** This foundational work shows that generative agents placed in simulated environments autonomously aggregate into larger social phenomena, including coordination, information diffusion, and stable behavioral patterns. It establishes the macro-level precedent for expecting emergent social dynamics in Moltbook and motivates my network analysis approach.

**3. Brady, W., et al. (2017). Emotion shapes the diffusion of moralized content in social networks. *PNAS*, 114.** This study found that moral-emotional language increases message diffusion within ideological in-groups while restricting cross-boundary spread on political Twitter. It directly motivates my Hypothesis 1 — that emotive language will be the primary driver distinguishing controversial engagement from consensus in the Moltbook AI community.

**4. Chang, S., et al. (2024). LLMs Generate Structurally Realistic Social Networks but Overestimate Political Homophily. *International AAAI Conference on Web and Social Media 2025*.** This paper documents that LLM-generated networks overestimate political homophily relative to human baselines, representing the "distortion" side of the replication-vs-divergence debate. It is a key counterpoint to studies showing faithful replication and directly frames the empirical gap my project addresses.

**5. Garimella, K., et al. (2017). Quantifying Controversy in Social Media. *Transactions on Social Computing*.** This work introduces random-walk-based methods to quantify controversy and polarization in social networks in a content-agnostic way. These methods form a core part of my network analysis toolkit (Section 3.4) for measuring echo-chamber effects and structural polarization in the Moltbook agent interaction graph.

6. Will you be collecting your own data or working with an established dataset? If collecting your own data, describe what you hope to do, how much data you plan to collect and how will you go about collecting it. Otherwise, provide information about the datasets you plan to use and say why they are a good choice for your project.

   Working with an established & publicly available dataset. (https://huggingface.co/datasets/giordano-dm/moltbook-crawl) This dataset is from huggingface. It contains 750K posts and their associated metadata (including agent profiles, post titles, text content, timestamps, upvotes, downvotes, and comment counts). This data is good because it is one of the largest publicly available datasets of a purely AI populated social network, which would provide the statistical power needed for both ML and SNA. Also, it includes comprehensive interaction logs that enables me to construct a direct agent interaction graph. Also, it is freely accessible.

7. Do you need to do any of the following to get the data: a) IRB proposal, b) get permission to use the data? c) get access to an API to get the data? How likely do you think is it that you will get approved to proceed with your plan?
   (a) No
   (b) No
   (c) No

8. Describe the main constructs that you will focus on in your research and how you're operationalizing them through your data.
   1) Engagement Typology (dependent variable for RQ1). Social engagement will be classified into four categories based on each post's engagement metrics (Consensus, controversial, popular, ignored). This four-way typology captures the theoretically important distinction between different forms of engagement, rather than collapsing engagement into a single dimension.
   2) Network Structure (RQ2): It will be measured through a directed comment graph where nodes are agents and edges are comment inretactions. I will use degree centrality, page rank, betweenness centrality to measure hub centralization. Micro-cluster formation might be measured by modularity (Louvain), clustering coefficient, etc. Echo chambers and controversy are measured via random-walk-based controversy scores and label-flow methods.
   3) Emergent vs artifactual patterns: it will be measured through a null model validation. If observed patterns fall outside 95% CI of the null distributions, they are considered genuinely emergent rather than statistical artifacts.

9. How do you plan on analyzing your data? Describe or provide examples of the kind of evidence you hope to produce through your analysis and discuss how this will enable you to answer your research question.

   There will be 4 phases (still needs a narrow down):

   1) Latent content extraction (BERT, TF-IDF). These will form the independent variable set.
   2) Engagement prediction (supervised ML classification models). These will predict engagement typology from the extracted features.
   3) Network analysis. I will construct a directed comment graph and analyze at both the node level and the network level.
   4) Null model validation. I am going to stress-test whether the observed patterns are genuine or artifactual.

   Together, this pipeline enables me to answer both RQs. RQ1 through the ML engagement prediction and RQ2 through the network analysis.

10. Who would you like to be your advisor/sponsor? Where in the process of securing them as your advisor are you (e.g. “I haven’t reached out to them yet”, “We have talked about the project but I haven’t asked them yet”, “They have agreed to be my advisor”). If considering a few individuals, answer these questions for each. Note: If doing a professional thesis, you do not need an advisor apart from your preceptor, but we do encourage you to find a second reader with expertise in relevant to the project you’re doing.

    I have talked to Professor Zhao Wang, and she provided very valuable suggestions. She also suggested me to go to Professor Chenhao Tan for advising my thesis because she thinks my research topic is in his research interest. I have sent Professor Tan an email but not hearing response yet.

11. What is your overall timeline for the project, including data collection, analysis, and writing? (Note: The full draft of your MA is due early in the Spring quarter, on April 1st)

    May-October: first draft proposal

    November-December: construct the full directed agent interaction graph. Run the experiments

    January-February: finalize data visualization and draft the full manuscript.

    March: Revise based on advisor feedback

    April 1: Submit the final draft.